

KB — Lamps

What the light source has to do for this instrument, how to judge one, and what the roster actually measures. Written 2026-09-04 around the Yuji-vs-halogen comparison, which is the first measurement able to separate the lamp from the instrument.

Companions: `D0C_lamp_410_680.md` (which Avonec LED combination to buy), `SPEC_lamp_rebuild.md` (the 7-emitter board), `KB_led_and_oil_spectra.md` (Avonec SPD sources), `KB_spectroscopy_physics.md` §7 (the instrument), `SPEC_capture_quality.md` §16 (the error budget). This note is the **general** one: it holds what is true of *any* lamp on this bench, so the three lamp-specific documents do not each re-derive it.

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1 • What a lamp is for here, and why "bright" is the wrong word

The instrument measures **transmittance**, $T = S/R$ — a sample burst divided by a reference burst through the same optics. Two consequences run through everything below.

The lamp cancels — until it doesn't. Any wavelength-dependent factor common to `S` and `R` divides out: the lamp's own spectral shape, the grating efficiency, the sensor QE. So a lamp does **not** need to be flat, or white, or high-CRI. This is the single most common wrong instinct about the light source, and

`SPEC_capture_quality.md §16.26` measured what actually does *not* cancel — re-seating the jar between the two bursts, at 1.7 % median against a 0.42 % instrument floor.

What a lamp must supply is DYNAMIC RANGE, per band, at the wavelengths the metric reads. The quantity that matters is not radiance but the digital number the camera lands on after the sample has absorbed. $T = S/R$ is a ratio of two 8-bit numbers, so:

where the reference sits	what happens to T
near 255 DN	the reference clips; T reads high and the band is silently dead
~50–200 DN	the working window
~16–50 DN	usable; one code step is 1–4 % of the level (<code>SPEC_capture_quality.md §16.23.10f</code>)
below ~16 DN	one code is >4 % before the gamma decode and ~2× that after; absorbance saturates as S hits 0

★ **The lamp's job is to put every band the metric reads into the middle of that table, simultaneously.** A lamp that is superb at 550 nm and dead at 630 nm is a bad lamp for this instrument even if it is brighter overall than one that is mediocre everywhere.

⚠ **And the bands are not equally forgiving.** `SPEC_capture_quality.md §16.24` measured the error budget as **17× asymmetric**: a given relative error in the Q band moves the metric ~17× more than the same error at the Soret. So deficiency in 560–630 nm costs far more than the same deficiency at 450 nm.

2 · The lamp families, and what each one is good for

family	continuum?	strength	🚫 the catch
Phosphor white LED (Yuji, Sansi)	yes, but a blue pump spike + phosphor hump	efficient, cool, cheap, stable	falls away past ~620 nm; the cyan dip at 470–510 is structural
Discrete LED board (Avonec 3 W)	no — a sum of narrow emitters	put light exactly where the bands are	ripple between emitters; <code>DOC_lamp_410_680.md</code> is the whole study
Halogen / incandescent	★ yes, and <i>analytically</i> — a Planck continuum	smooth, gap-free, rises monotonically into the red	hot, power-hungry, blue-poor, ages
CFL / fluorescent	🚫 no — narrow lines on a weak continuum	★ the wavelength standard (Hg 404.66 / 435.83 / 546.07 / 576.96 / 579.07, Eu ³⁺ 611.6 / 626.6 / 650.7)	🚫 ~17× dimmer per unit exposure than the Sansi — cannot be the measurement lamp (<code>KB_spectroscopy_physics.md §4.1</code>)

★ **Two different jobs, and they do not have to be the same lamp.** The **measurement** lamp needs dynamic range in the bands. The **calibration** lamp needs known, resolvable lines. The CFL is disastrous at the first and unbeatable at the second; the project already uses it that way.

2.1 ★ The property that made this document possible: a halogen is *analytically known*

A tungsten filament is a **grey body**. Its emission is Planck's law times an emissivity that falls from about 0.45 at 450 nm to about 0.42 at 700 nm — a few percent across the entire visible range. So its spectral power distribution is **known in closed form from one number** (the colour temperature), and even that number barely matters in the red.

★ That is a capability no other lamp on the roster has, and §4 spends it: **dividing a measured halogen frame by Planck leaves the instrument response**. Every previous attempt to characterise the red end was blocked precisely because the lamp's own SPD was unknown exactly where the question was —

KB_spectroscopy_physics.md §7.2 states the block outright: "*IR-cut filter + sensor QE + the source's own decline — one lamp cannot separate the three.*"

3 · The measurement: Yuji SunWave LED vs a 60 W halogen

Two frames from *Settings > Development > Capture images* (2026-09-04), same session, same optics, same ROI box, camera ELP 32e4:8830. Neither frame saturates (peaks 217 and 234 DN), so each lamp is exposed to fill the sensor at its own maximum — a like-for-like comparison, not an exposure artefact.

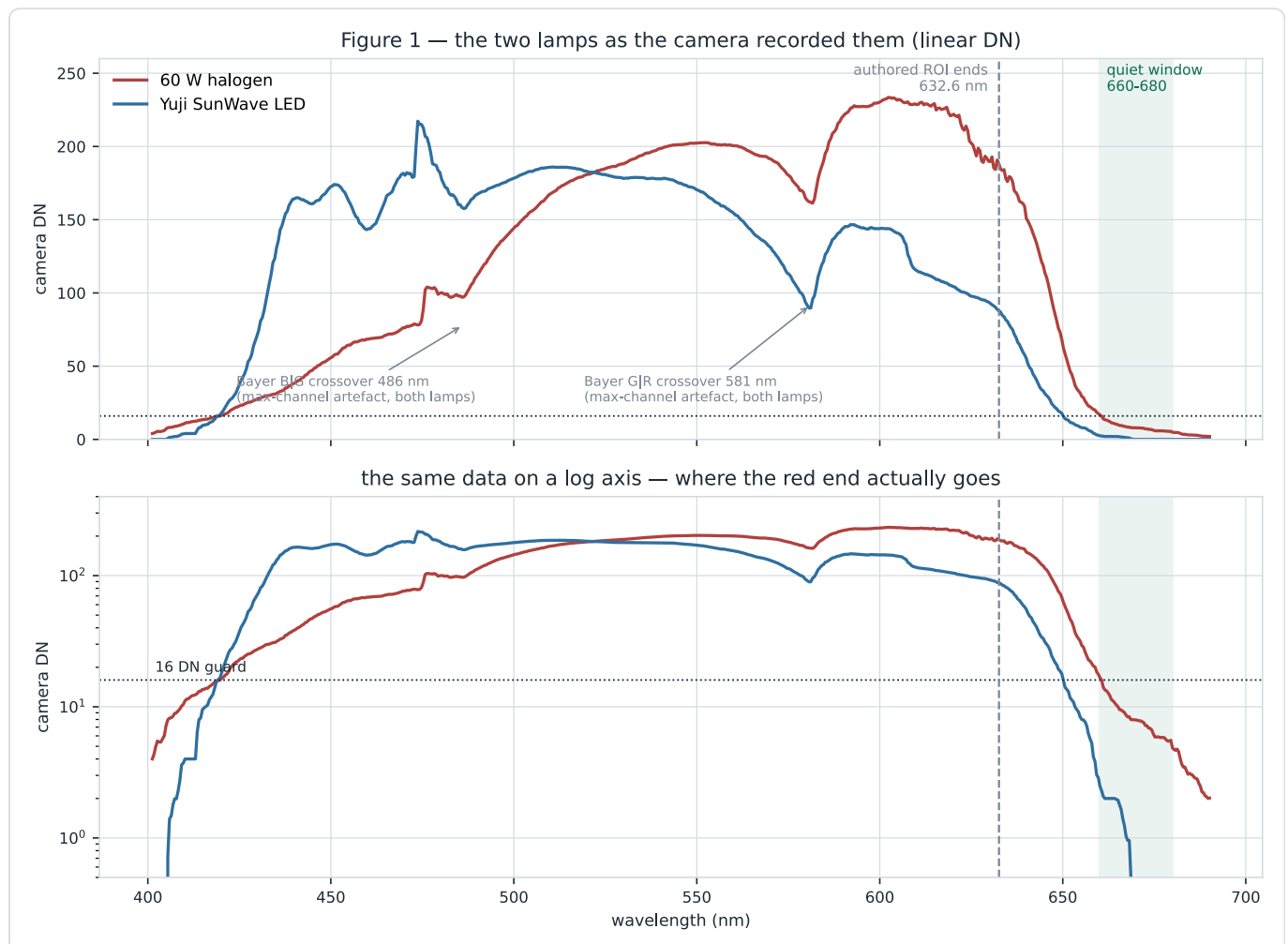


Figure 1 — the two lamps as the camera recorded them, lamp × instrument together. Top: linear DN. Bottom: the same data on a log axis, which is the only way to see what the red end does. The dashed vertical is where the *authored* ROI ends (632.6 nm) — everything right of it is invisible to the measurement pipeline today. The two grey arrows mark artefacts of the **max-channel** reduction, not features of either lamp.

3.1 The red edge, as measured

level	Yuji SunWave LED	60 W halogen
≥ 50 DN	640.5 nm	651.6 nm
≥ 20 DN	648.4 nm	658.8 nm
≥ 16 DN (the guard)	649.7 nm	660.2 nm
≥ 10 DN	652.9 nm	665.1 nm
≥ 5 DN	657.9 nm	679.5 nm
≥ 2 DN	661.1 nm	690.1 nm (the raster edge)
exactly 0	668 nm	never, inside the raster

⇒ The halogen buys about 10 nm at the working floor and reaches the raster edge at 2 DN. Both lamps are dead well before 700 nm.

3.2 △ Two notches that belong to the camera, not to either lamp

The pipeline reduces each column by **max-channel** (`SpectralColorUtil.toGrayMaximum`, chosen in §15 because `qGray` suppresses blue). Where two Bayer channels hand over, the maximum of the three is at its lowest, so max-channel imprints a **notch at every crossover**. Measured here, on both lamps:

crossover	Yuji	halogen
B = G	486.2 nm	486.2 nm
G = R	581.0 nm	581.5 nm

★ These double as a wavelength-scale check that needs no calibration lamp. The crossovers are a property of the sensor's colour filters; two lamps of entirely different families agreeing on both, to 0.5 nm, pins the scale used throughout this note. The 581 figure is the one already on record — it is why `KB_spectroscopy_physics.md` §4.1a reads the Q band at 568 nm rather than 574.

△ Do not read a notch at 486 or 581 nm as a lamp feature or a pigment feature. It is the reduction.

4 · ★★ Dividing out the source: the ELP has a hard edge at 642 nm

Divide the measured halogen by Planck and what is left is the instrument — grating, lens, IR-cut filter, Bayer dye and silicon QE, all together. Normalised at 600 nm:

△ "The instrument" here means the ELP path specifically — the bench/dev unit, and the one the whole 98-run archive was captured through. §6.1 shows the production camera does **not** share its edge.

nm	2700 K	2900 K	3100 K
550	1.0921	1.0333	0.9846
600	1.0000	1.0000	1.0000
620	0.8158	0.8323	0.8470
630	0.5455	0.5618	0.5764
640	0.3134	0.3256	0.3367

nm	2700 K	2900 K	3100 K
650	0.0423	0.0444	0.0463
655	0.0079	0.0083	0.0087
660	0.0022	0.0024	0.0025
670	0.0004	0.0004	0.0005

★ **Colour temperature is irrelevant to the conclusion** — under 10 % spread across 2700–3100 K in the red, because Planck is smooth and slowly varying there. (It matters in the blue: the 450 nm entry moves 46 % across the same range. Nothing here depends on the blue.)

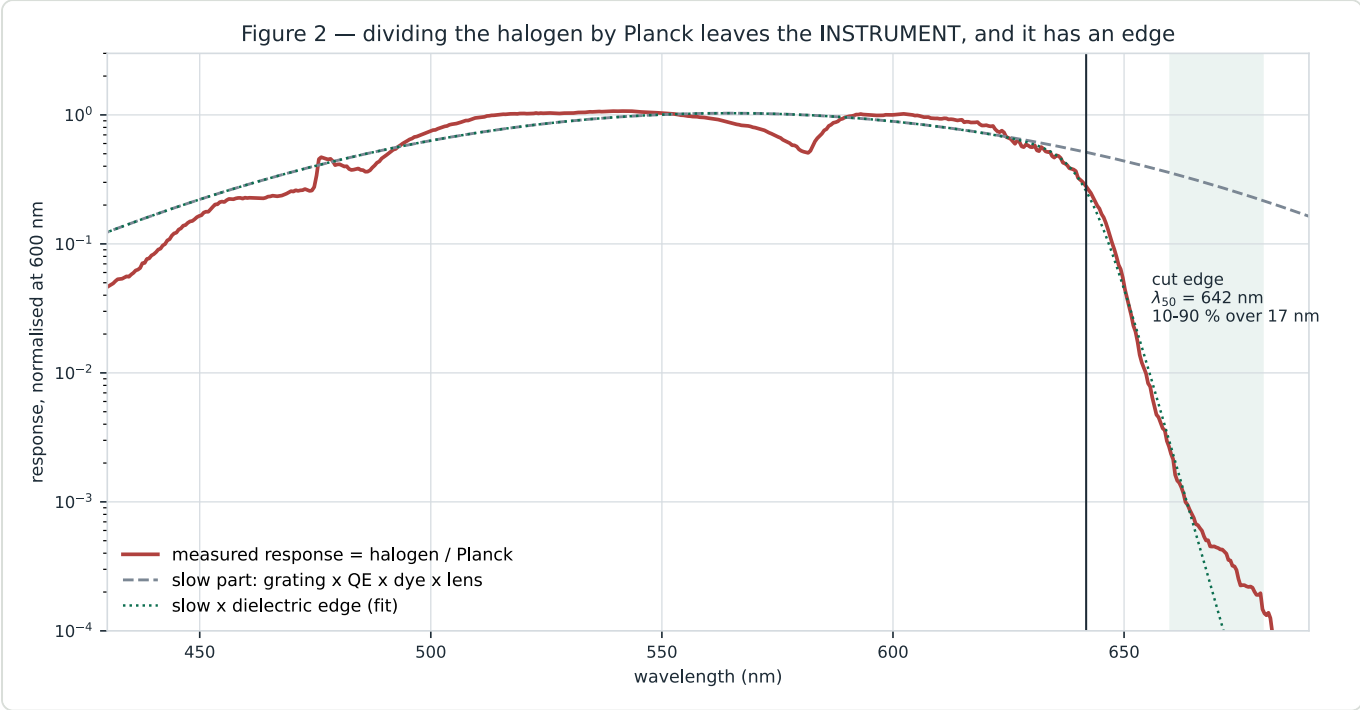


Figure 2 — dividing the halogen by Planck leaves the instrument, and the instrument has a cliff. The measured response (red) tracks a slowly-varying curve to about 635 nm and then falls off an edge that no smooth optical component can produce. The dashed grey line is the fitted slow part — grating × QE × dye × lens; the dotted green is that slow part times a fitted dielectric edge.

4.1 The steepness is the whole argument — stated without a model

span	response lost	pace
450 → 550 nm	<i>rising</i>	—
550 → 620 nm	0.09 decades	10× every 745 nm
620 → 640 nm	0.41 decades	10× every 49 nm
630 → 650 nm	1.10 decades	10× every 18 nm
640 → 660 nm	2.14 decades	10× every 9 nm

★ **The red edge is about 50× steeper per nm than anything else in the response.** Silicon quantum efficiency, Bayer dye transmission and grating efficiency all vary over *hundreds* of nanometres; none of the three can lose a factor of ten in nine. A **dielectric interference filter** — an IR-cut / hot mirror — is the only component in the chain with an edge of that shape.

Fitting `response = exp(quadratic) × 1/(1+exp((λ-λ50)/w))` puts the edge at:

λ₅₀ = 641.8 nm, w = 3.76 nm — 10 % to 90 % over 16.5 nm.

★ And it does not move: **641 ± 1 nm** under `pow1.8` / `pow2.2` / `pow2.4` / sRGB decoding and across 2700–3100 K. The 10 → 90 width ranges 15–22 nm over the same set. The one number the fit is sensitive to is the decode exponent, and it moves the *width*, not the position.

4.2 So what exactly is owed to the IR-cut filter?

Splitting the total into the slow part and the edge, relative to 620 nm:

nm	total attenuation	of which slow	of which the edge	the edge's share
630	×1.5	×1.15	×1.0	23 %
640	×2.6	×1.36	×1.6	61 %
650	×18.8	×1.64	×10.3	82 %
655	×100.4	×1.82	×36.1	86 %
660	×353.2	×2.03	×132.5	87 %
670	×1875	×2.55	×1771	89 %

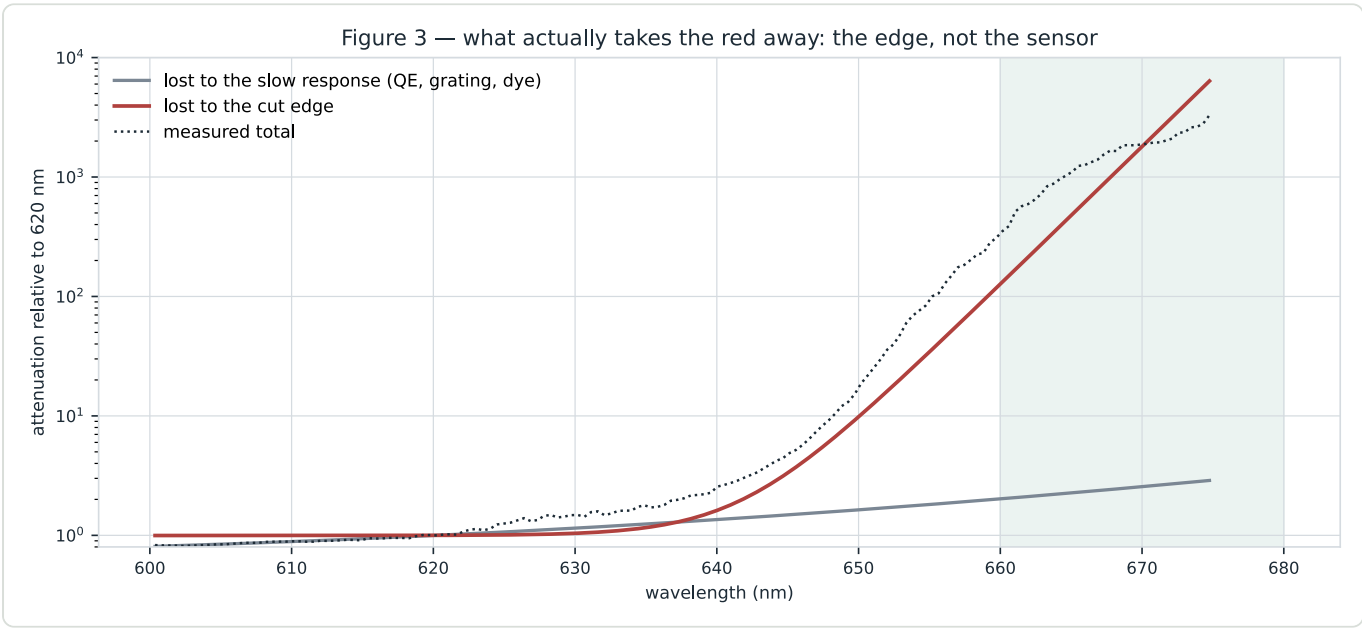


Figure 3 — what actually takes the red away. Everything slow — quantum efficiency, grating, dye, lens — costs a factor of 3 across the whole span. The edge costs a factor of 6000. Beyond ~645 nm the measured total and the edge are the same curve.

⇒ **The answer, in one line: at 650 nm about 82 % of the loss is the filter and at 660 nm about 87 %; everything else in the camera and optics together costs less than a factor of 2 across 620–660 nm.**

⚠ **Read the split honestly.** What is *measured* is the total. The decomposition into "slow × edge" is a **model**, and the share column inherits that model's assumptions. What is robust is the thing the model was fitted to and the thing that motivates it: the log-slope changes by ~50× within 30 nm, which no smooth component explains. **➖** This is **not** a direct measurement of the filter; that needs a de-filtered camera (\$7).

4.3 Three objections, answered

"It is the quantisation floor, not a real edge."  No. The whole 630 → 660 collapse happens between **190 DN and 17 DN** on the halogen — every point is above the 16 DN guard and far above any floor. The collapse is measured on well-exposed data; only past 665 nm does quantisation take over, which is why the fit stops at 668.

"It is the halogen bulb's own coating." IRC (infrared-reflecting) halogen capsules exist, but their edge is in the **near infrared, 700–800 nm**, not at 642 nm — and a coating cutting at 642 nm with a 16 nm width would make the bulb visibly cyan. The recorded halogen looks like a normal warm source, with its measured maximum at 602 nm.

"It is the grating." A transmission grating's efficiency varies smoothly over hundreds of nanometres, and there is no mechanism in the geometry for a sharp cut. It is part of the "slow" column.

5 · What this overturns

5.1 `DOC_lamp_410_680.md` §6's withdrawal must be re-opened

§6 was withdrawn on 2026-08-07 on the grounds that its evidence (§7.2's "40× between 631 and 657 nm") came through a **CFL**, whose red output is **Eu³⁺ line** emission — so the 40× was that lamp's own spectrum, not the instrument's.  **That criticism was correct, and it does not apply here.** A halogen has no lines. Its continuum is known. The 350× between 620 and 660 nm cannot belong to the source, because the source provably *ris*es across that span.

⇒ The working assumption recorded at §6.1 — *"the optics deliver to ~700 nm and the camera captures it"* — **is refuted for anything past ~650 nm.**

5.2 §6.2a's open question is decided, without the europium lines

§6.2a set the test explicitly: *"Lines visible ⇒ the camera passes 690 nm, §6's withdrawal stands, and the collapse belongs to the lamps. Lines absent ⇒ the IR-cut is the gate and §6 must be re-opened."*

 **The halogen answers it without needing to resolve a line**, because a continuum whose shape is known does the same work as a line whose position is known. **The IR-cut is the gate.** Marking `EUROPIUM_RED_FAR_680/690/700` is still worth doing as an independent check, but it is no longer the blocker.

5.3 An independent replication of `SPEC_capture_quality.md` §16.28.4

§16.28.4's run `20260808B` (Yuji, ROI opened to 690 nm, 2026-08-09) is reproduced by this frame — a different session, a different exposure, the same lamp:

nm	§16.28.4	this frame	ratio
630	39.70	27.61	1.44
640	11.50	8.85	1.30
650	0.70	0.557	1.26
656	0.13	0.115	1.13
660	0.01	0.010	1.05

★ **The same shape across 4000× of dynamic range, to within one constant factor.** §16.28.4's Yuji row is confirmed, and the disagreement recorded there is therefore entirely on the Sansi side.

5.4 🚫 And it kills one of §6.2a's three reconciliations

§6.2a offered three ways to reconcile "the Sansi still returns 115 DN at 656 nm" (§16.25.4) with the collapse: **(1)** the V1 Sansi genuinely owns the deep red, e.g. via a KSF/PFS line phosphor; **(2)** the red end of that figure is mis-scaled; **(3)** its mid-range is clipped.

At a response of 0.00776 relative to 620 nm, a lamp reading 115 DN at 656 nm would have to **emit about 30× more at 656 nm than at 620 nm** (exposure cancels in the ratio). 🚫 No phosphor does that — a KSF/PFS Mn⁴⁺ line emitter included, and §6.2a already notes the 2026-08-09 runs show **no line structure** at 648 or 660 on either lamp.

⇒ **Reconciliation 1 is dead. 2 and 3 survive**, and 2 (a wavelength axis transferred from a screenshot ending near 676 nm) is the one this note would bet on.

6 · What it means for the red extension

The metric family's binding constraint is the **630 nm clamp** — the authored ROI ends at 632.6 nm, and `spectracs-metric-family-2026-08-21` records eight independent arguments for pushing it redward. This measurement adds a ninth and sharpens the other eight, because it is the first one made on the actual lamp rather than inferred.

What is free today. The halogen holds **190 DN at 630 nm and 62 DN at 650 nm**. Re-authoring the ROI from 632.6 nm out to ~655 nm costs nothing in signal-to-noise with hardware already on the bench. ⚠️ That is a *photometric* statement only; §7.13's starved-regime warning (a concentration-dependent compression, not merely noise) applies to any new band and has to be re-checked on oil, not on an empty beam.

What is not free — ⚠️ on the ELP. The **660–680 nm quiet window** — the first pigment-free region, and the baseline anchor the metric has never had (`DOC_lamp_410_680.md` §2.2) — sits *behind the edge*. The halogen delivers 17 DN at 660 and 5 DN at 680. 🚫 **No lamp change reaches it on this camera.** The Avonec deep-red star, the R2 board, a brighter white: all of them are multiplied by the same 0.0024. It converts "buy a redder lamp" from a plan into a dead end — ⚠️ **for the ELP bench unit.**

★★ 6.1 🚫 AND THE EDGE IS THE ELP'S, NOT THE INSTRUMENT'S — the production camera has no IR-cut (*Edwin, 2026-09-04*)

⚠️ **Correction to the paragraph above, made the same day it was written.** Everything in §4 was measured on the ELP `32e4:8830` — the bench/dev unit. The camera intended for the **production batch** is the **Microdia/Sonix** `0c45:6366` (`KB_spectroscopy_physics.md` §7), and Edwin ran the remote test on it:

a TV remote in the dark shows **a white dot with a purple/violet halo**.

★ **That halo is the diagnostic, not just the dot.** A silicon sensor's three Bayer dyes all become transparent in the near infrared, so an unfiltered sensor renders an 850/940 nm source as white with a violet fringe — every channel responding at once. A camera *with* an IR-cut returns nothing, or a dim dot with no colour fringe. ⇒ **the production camera has no effective IR block.**

What this changes, and what it does not:

— changes	"660–680 nm is unreachable" is a fact about the ELP , not about the instrument concept. §7's de-filtered camera is no longer a purchase — one is already in hand
— changes	the R2 board's deep-red half, and <code>D0C_lamp_410_680.md</code> §5.4/§7.3, are re-opened on the production camera. They stay dead on the ELP
★ does not change	every number in §4 and §5. They describe the ELP path, which is what the entire 98-run archive was captured through
⚠ does not follow	"no IR-cut" ⇒ "good 660–690 response". It removes the <i>edge</i> ; what remains (silicon QE, red dye) is near plateau there, so a good response is <i>expected</i> — but expected is not measured

⚠ **Two new risks that only appear once the filter is gone**, neither of which the ELP data can speak to: **stray NIR scatter** raising the floor across the whole band, and **second-order diffraction**. In a spectrograph most NIR disperses off the raster (§7), but *where* depends on this camera's own dispersion, which is not yet calibrated.

7 · ★ The decisive experiment, and what it predicts

— Everything above attributes by *shape*. One experiment measures it directly: **capture the same halogen frame through a camera with no IR-cut filter**. ★ Per §6.1 that no longer needs a purchase — **the Microdia `0c45:6366` already on the bench is one**.

★ The prediction is quantitative and falsifiable. Extrapolating the fitted slow term:

nm	halogen today	predicted, no IR-cut	gain
650	62 DN	~177 DN	×10
660	17 DN	~161 DN	×147
670	8 DN	~145 DN	×596
680	5 DN	~129 DN	×1341

⚠ **The slow term is extrapolated past 650 nm**, so these are predictions with real uncertainty in the *level* — though not in the *order of magnitude*, because silicon QE and the red Bayer dye are both near their plateau through 650–700 nm. If a de-filtered camera returns 650–680 nm at anything like these levels, §4's attribution is confirmed and the quiet window opens with the lamp already on the bench.

★ **And a spectrograph is the rare instrument that can afford to lose its IR-cut filter**. In a camera, removing it floods every pixel with near-infrared. Here the NIR is **dispersed** — 700–1000 nm lands on columns past 2668, which are off the 2592-wide raster entirely. ⚠ The two risks that remain are **stray NIR scatter** raising the floor across the whole band, and **second-order diffraction** (2nd order of 340 nm lands where 680 nm does), the latter largely handled by glass absorbing the UV.

★ 7.1 How to run it, in order of what it costs

★ **Step 1 needs no calibration at all, and answers the question**. Point the halogen at the Microdia in the dev capture view and look at the red end of the raw frame for a **cliff**. A 16 nm-wide collapse is scale-free — you do not need nanometres to see that it is or is not there. ⇒ *No cliff = no filter edge = §6.1 confirmed and the quiet window is reachable on the production camera*.

Step 2 puts numbers on it, and the established route is already in the record: a **CFL frame**, then the mercury anchors **435.83 / 546.07 nm** for a two-point linear px → nm (`KB_spectroscopy_physics.md` §7.2 did

exactly this at 0.5057 nm/px). Then divide the halogen by Planck as in §4 and compare the response against the ELP's.

△ **Two things block a same-day run and should be budgeted:**

1. 🚫 `CaptureBackend` **hardcodes 2592×1944** — the ELP's calibration resolution — with its own `TODO: make this per-sensor ... when a second camera lands`. The Microdia's native modes are not that, so the pin has to become per-sensor first, or the frame comes back snapped to something else.
2. 🚫 The Microdia has **no calibration profile** (the second `spectrometer_calibration_profile` row is all NULL). Step 1 does not care; step 2 needs the CFL run above.

△ And a caution on reading the result: an unfiltered camera's own red response is **not** the ELP's slow term. §4's quadratic was fitted through the ELP's optics and dye set; it may not transfer. Compare the two cameras' *shapes*, not their absolute levels.

8 · How to characterise any new lamp on this bench

The recipe this note followed, in order — it takes one evening and no new hardware:

1. **Capture it in *Settings > Development > Capture images***, exposure set so the peak lands at 210–240 DN. △ Check it does not clip: a clipped mid-range makes the red *shape* uncomparable, which is exactly what damaged §16.25.4's Sansi row.
2. **Read the ROI box as the EXTENDED ROI** (400.0–690.8 nm), not the authored one. They differ by 58 nm of red and the box on screen is the extended one.
3. **Check the scale on the Bayer crossovers** — B=G at 486.2 nm, G=R at 581 nm. Any lamp, no calibration needed. If they land elsewhere, the ROI or the cubic is wrong; stop.
4. **Report DN at the metric's bands**, not a normalised curve. Normalising to the peak hides the only property that matters (§1).
5. **If it is a thermal source, divide by Planck** and you get the instrument for free.
6. 🚫 **Never compare two lamps' red ends at different exposures without saying so** — half of the contradictions in the record trace to that.

Reproduce everything here with:

```
./venv/bin/python diagnostics/lamp_yuji_vs_halogen.py --figures
```

9 · What would change these conclusions

- ★★ **The Microdia halogen frame (§7.1) showing no cliff** ⇒ §4.2's attribution confirmed and the quiet window opens **on the production camera**. This is the cheapest open experiment in the note and it needs no purchase — it is the one to run first.
- **The Microdia returning the same collapse** ⇒ 🚫 the edge is not the ELP's IR-cut; §4 is wrong and the next suspects are the grating block and the M12 lens coating, which the two cameras share.
- **The Microdia's own red response falling for a different reason** (its dye set, its lens) ⇒ △ neither confirms nor refutes §4; it would mean the production camera needs its own characterisation, not that the ELP's edge was misattributed.
- **A raw (undecoded, non-preview) frame of the same halogen** ⇒ removes the last assumption in §4, the gamma decode. Expected to move the *width* by a few nm and the position by ~1 nm.

- **The Eu^{3+} 687.7 / 693.7 / 707.0 nm lines appearing** $\Rightarrow$  would contradict §4 outright and re-open §6.2a. On these numbers they should be invisible: the response at 690 nm is $\sim 1 \times 10^{-4}$.
- **A second halogen bulb of a different make** $\Rightarrow$ rules out the IRC-coating objection (§4.3) by construction rather than by argument.